

PROJECT AYA

Compression, Equity, and the Architecture of Linguistic Inclusion

Cohere Labs × Wayy Research · Buffalo, NY · Est. 2026 · Open Science / Open Weights

Multilingual	Mamba SSM	Distillation	70+ Languages	MoE	Open Source	Edge AI
--------------	-----------	--------------	---------------	-----	-------------	---------

01 — Overview & Problem Statement

The Problem

Most efficient AI models are English-centric. The architecture of language technology is leaving billions behind.

70%+	of internet users are non-English speakers, yet most edge-deployed models serve English only
$O(n^2)$	Transformer attention cost makes multilingual models impractical on consumer hardware and edge devices
0	Multilingual Mamba large language models exist today — this gap is the opportunity

Core Research Question

Can transformer-based multilingual models be distilled into Mamba architectures while preserving multilingual capability, structured tool use, and cross-lingual reasoning — and does the compression degrade uniformly across language families?

02 — Architecture: Aetheris (Hybrid Mamba-MoE)

A novel student architecture combining selective state spaces with sparse mixture-of-experts, distilled from Cohere's Tiny Aya.

Teacher → Student



3-Stage MambaInLlama Pipeline

1	Layer Alignment	Map transformer attention layers to Mamba SSM blocks with structural correspondence
2	KL Distillation	Soft-target training with KL divergence to transfer knowledge from teacher to student
3	Supervised Fine-Tuning	Restore multilingual capability and structured tool calling via targeted SFT

Aetheris Design Details

SSM Blocks	Ψ	O(n) selective scan with constant memory. Mamba backbone enables linear-time sequence processing
Sparse MoE	Σ	4 expert FFNs with top-1 routing and load balancing. Only one expert fires per token.
Hybrid Design	Δ	SSM blocks on odd layers, MoE on even layers. 24 total layers with weight-tied embeddings and gradient checkpointing.

03 — Evaluation Scope: 10 Languages, 5 Families

Evaluation spans diverse typologies — analytic, agglutinative, fusional, polysynthetic — to measure whether compression is equitable.

Language	Family	Script	Typology
English	Indo-European	Latin	Analytic
Spanish	Indo-European	Latin	Fusional
Hindi	Indo-European	Devanagari	Fusional
Mandarin	Sino-Tibetan	CJK	Analytic/Isolating
Arabic	Afroasiatic	Arabic	Fusional/Semitic

Language	Family	Script	Typology
Swahili	Niger-Congo	Latin	Agglutinative
Turkish	Turkic	Latin	Agglutinative
Japanese	Japonic	CJK+Kana	Agglutinative
Indonesian	Austronesian	Latin	Analytic
Telugu	Dravidian	Telugu	Agglutinative

04 — Evaluation Benchmarks

Benchmark	Description
mGSM	Multilingual grade school math across 10 languages, 250 problems each. Tests arithmetic reasoning across scripts.
XCOPA	Cross-lingual commonsense reasoning across 11 languages. Causal inference beyond English-centric assumptions.
Tool Calling	Structured JSON generation and function selection in 10+ languages. Nearly unstudied in multilingual settings.
CPU Throughput	Tokens/sec, peak memory, time-to-first-token on consumer hardware. Core practical objective.

Degradation Equity Score

Key novel metric: the variance of accuracy drop across language families after distillation. Measures whether compression picks winners and losers. Lower is better — compression should not pick winners.

05 — Technology Stack

ML / Training	Infrastructure
- PyTorch - HuggingFace Transformers - HuggingFace Datasets - Mamba SSM - CUDA	- Docker + GitHub Actions - RunPod A100 - FastAPI + SSE Streaming - llama.cpp (edge deployment) - Pydantic

06 — Project Roadmap & Milestones

Phase	Name	Key Deliverables	Target
01	Baseline	Load Tiny Aya, establish CPU tokens/sec in 5 languages. Validate tokenizer compatibility. Set performance baseline the distilled model must beat.	Q1 2026
02	Distillation	Execute 3-stage MambaInLlama pipeline on RunPod A100. Layer alignment, KL distillation, then multilingual SFT with tool calling recovery.	Q2 2026
03	Evaluation	Full benchmark suite across 10 languages. Equity analysis, ablation studies, and paper preparation targeting EMNLP / ACL / NeurIPS.	Q3 2026

07 — Team Work Tracker

Use this section to assign tasks, track progress, and manage blockers across phases.

Phase	Task	Status	Owner	Due Date
Phase 01	Environment setup & dependency validation	Not Started	TBD	
Phase 01	Load Tiny Aya, validate tokenizer compatibility	Not Started	TBD	
Phase 01	Establish CPU baseline (tokens/sec) in 5 languages	Not Started	TBD	
Phase 01	Document baseline benchmark results	Not Started	TBD	
Phase 02	Layer alignment: attention → Mamba SSM blocks	Not Started	TBD	
Phase 02	KL distillation training on RunPod A100	Not Started	TBD	
Phase 02	Supervised fine-tuning: multilingual + tool calling	Not Started	TBD	
Phase 02	Validate Aetheris student model quality	Not Started	TBD	

Phase	Task	Status	Owner	Due Date
Phase 03	Run mGSM benchmark (10 languages)	Not Started	TBD	
Phase 03	Run XCOPA benchmark (11 languages)	Not Started	TBD	
Phase 03	Tool calling evaluation (10+ languages)	Not Started	TBD	
Phase 03	CPU throughput benchmarking	Not Started	TBD	
Phase 03	Compute Degradation Equity Score across families	Not Started	TBD	
Phase 03	Ablation studies	Not Started	TBD	
Phase 03	Paper draft for EMNLP / ACL / NeurIPS submission	Not Started	TBD	
Phase 03	Open-weight model release	Not Started	TBD	

Status Key

Not Started	Work has not yet begun
In Progress	Actively being worked on
Complete	Done and verified

08 — Meeting Notes & Decisions

Add meeting notes, key decisions, and action items below.

Date	Decision / Note	Action Items
3/1/2026	Added comments for improving this document, specifically addressing issues (1) mapping transformer blocks to mamba, and about developing inexpensive benchmarks for the linguistic reasoning and tool use. (1) Transformer to Mamba In...	Have full meeting tomorrow to discuss what development cycles are needed to achieve these goals. Aetheris_Design_Document.docx I
3/2/2026	Talked about the involvement in the project and aligned new folks to the research problem.	Based upon the literature and Arjit's comments, the CKA is a bottleneck but these studies have been done previously and might not reflect our architecture/pipeline and should be taken with a grain of salt. I recommend experimenting first and looking at the evaluations to see if the CKA is indeed a critical bottleneck for threshold < 0.75 Centered Kernel Alignment (CKA) fo...
3/3/26	I mean you gotta imagine not only do we have to find a way to do a compute efficient transfer of information from transformers to the mamba architecture, we have to get tiny aya compatible results on their bench. If you have access to deep research throw this in to there and see what novel or unique methods we can do to essentially hyper efficiently distil a transformer architecture to mamba, that alone is a phds worth of research we'll all do in	

Date	Decision / Note	Action Items
	about a day, side publishing rights first named author to whoever cracks this Once we have a methodology we can build an architecture, experimentation loop, and run some training jobs. Might have to get scrappy with it, I have some run pod credits and some Hetzner vms i m ight spin up a machine for research you can ssh into. Let's see what we come up with for research and we'll go from there	

Project Aya · Multilingual Transformer-to-Mamba Distillation · CC-BY-NC · Open Science. Open Weights. Open Questions.